

JEE Main Physics – 10 Questions

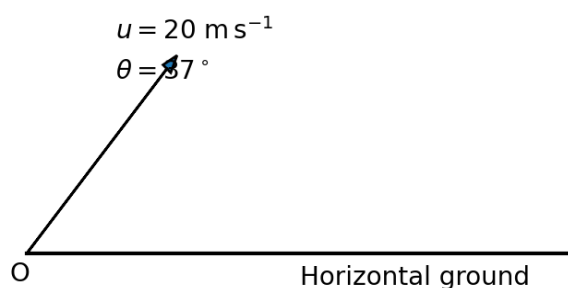
Questions include numerical problem-solving, multi-step equations, and selected diagrams.

Q1. Projectile Motion

A particle is projected from point O on horizontal ground with speed $u = 20 \text{ m s}^{-1}$ at an angle 37° above the horizontal, as shown. Take $g = 10 \text{ m s}^{-2}$ and use $\sin 37^\circ = 3/5$, $\cos 37^\circ = 4/5$.

Find:

- (a) the time of flight,
- (b) the horizontal range, and
- (c) the maximum height reached by the particle.



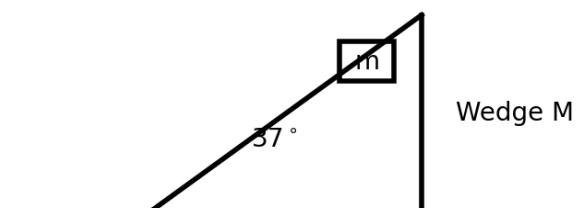
Q2. Block on a Smooth Wedge

A small block of mass $m = 2 \text{ kg}$ is placed on a smooth wedge of mass $M = 8 \text{ kg}$ and angle 37° , as shown. The wedge rests on a smooth horizontal surface.

Find:

- (a) the acceleration of the wedge,
- (b) the acceleration of the block with respect to the wedge, and
- (c) the horizontal acceleration of the block with respect to the ground.

Take $g = 10 \text{ m s}^{-2}$.



Q3. Work–Energy and Variable Force

A particle moves along the x-axis under a force $F(x) = (4x^2 - 6x + 2)$ N, where x is in metres. If the particle starts from rest at $x = 0$ and has mass 2 kg, find its speed when it reaches $x = 3$ m.

Use the work–energy theorem carefully:

$$W = \int_0^3 F(x) \, dx = \Delta K.$$

Q4. Gravitation

A satellite is moving in a circular orbit of radius $r = 3R$ around Earth, where R is the radius of Earth. Let g be the acceleration due to gravity at Earth's surface.

Find:

- (a) the acceleration due to gravity at orbit,
- (b) the orbital speed in terms of g and R, and
- (c) the time period of revolution in terms of g and R.

Q5. Thermodynamics

One mole of an ideal monoatomic gas is taken from state A to state B by heating at constant pressure $P = 2 \times 10^5$ Pa. The temperature increases from 300 K to 600 K.

Find:

- (a) the work done by the gas,
- (b) the change in internal energy, and
- (c) the heat supplied to the gas.

Use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ and for a monoatomic ideal gas:

$$\Delta U = (3/2)nR\Delta T.$$

Q6. Ray Optics

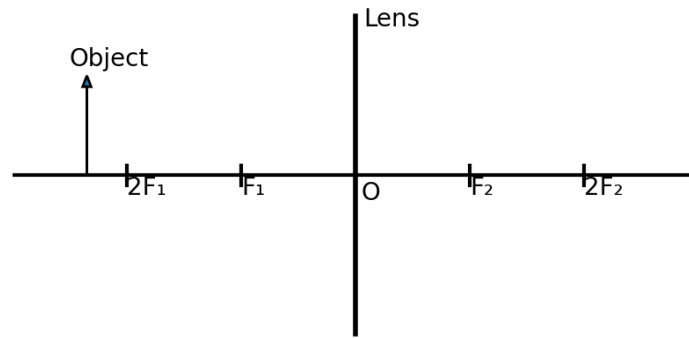
A convex lens of focal length 20 cm is shown. An object is placed 30 cm to the left of the lens.

Find:

- (a) the image distance,
- (b) the magnification, and
- (c) the nature of the image.

Use the sign convention and lens formula:

$$1/f = 1/v - 1/u.$$



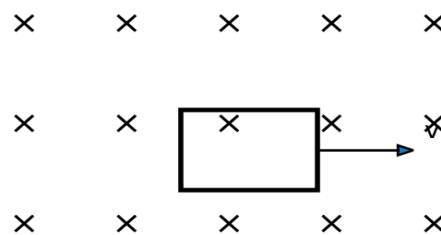
Q7. Electromagnetic Induction

A rectangular conducting loop of dimensions $20\text{ cm} \times 10\text{ cm}$ moves with speed $v = 5\text{ m s}^{-1}$ into a uniform magnetic field $B = 0.5\text{ T}$ directed into the plane of the paper, as shown. The 20 cm side is perpendicular to the velocity.

Find:

- the induced emf while entering the field, and
- the induced current if the resistance of the loop is $0.2\ \Omega$.

Uniform B into the page



Q8. Current Electricity / RC Circuit

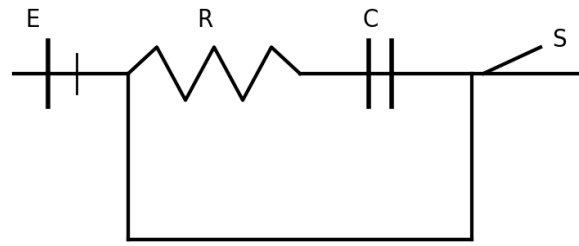
In the RC circuit shown, the switch S is closed at $t = 0$. The battery emf is $E = 12\text{ V}$, resistance $R = 3\text{ k}\Omega$, and capacitance $C = 200\ \mu\text{F}$.

Find:

- the time constant τ ,
- the initial charging current, and
- the charge on the capacitor after time $t = \tau$.

Use:

$$q(t) = CE(1 - e^{-(t/RC)}).$$



Q9. Modern Physics

Light of wavelength $\lambda = 400 \text{ nm}$ falls on a metal surface whose work function is 2.2 eV .

Find:

- (a) the energy of the incident photon,
- (b) the maximum kinetic energy of the emitted photoelectrons, and
- (c) the stopping potential.

Use $hc = 1240 \text{ eV}\cdot\text{nm}$.

Q10. Rotational Motion

A uniform rod of length $L = 1.2 \text{ m}$ and mass $M = 3 \text{ kg}$ is hinged at one end O and is free to rotate in a horizontal plane. A particle of mass $m = 1 \text{ kg}$ strikes and sticks to the free end of the rod with speed 6 m s^{-1} perpendicular to the rod, as shown.

Find the angular speed of the system just after collision.

Take the moment of inertia of the rod about one end as $I_{\text{rod}} = (1/3)ML^2$.

